Time–temperature scaling of isothermal sintering by vapor transport

Problem. At uniform temperature T, in the sharp-interface limit of the model, the vapor density ρ_v in the pore and the ice surface Γ (normal $\mathbf{n}$ into the pore, normal velocity $v_n > 0$ for growth, curvature χ) obey

$$\frac{\partial \rho_v}{\partial t} = D_v \, \nabla^2 \rho_v \qquad \text{in the pore,} \tag{1}$$

$$\rho_{\text{ice}} \, v_n = D_v \, \frac{\partial \rho_v}{\partial n} \qquad \text{on } \Gamma \text{ (conservation of water),} \tag{2}$$

$$\frac{\rho_v - \rho_{vs}^I}{\rho_{vs}^I} = d_0 \, \chi + \beta \, v_n \qquad \text{on } \Gamma \text{ (Gibbs–Thomson),} \tag{3}$$

with

$$d_0 = \frac{\gamma \, m}{\rho_{\text{ice}} \, k_B T}, \qquad \beta = \frac{1+\delta}{\alpha_c} \, \frac{\rho_{\text{ice}}}{\rho_{vs}^I(T)} \sqrt{\frac{2\pi m}{k_B T}}, \tag{4}$$

where $\delta = \Delta\beta/\beta_{\text{sub}} = 0.020$ is the thin-interface term. Temperature enters through $\rho_{vs}^I(T)$, $D_v(T) \propto T^{1.81}$, d_0 and β.

Scales. With $\bar{R}$ the mean grain radius,

$$\mathbf{x} = \bar{R} \, \mathbf{x}', \qquad \chi = \frac{\chi'}{\bar{R}}, \qquad \frac{\rho_v - \rho_{vs}^I}{\rho_{vs}^I} = \frac{d_0}{\bar{R}} \, u, \qquad v_n = \frac{d_0}{\bar{R}\,\beta} \, v', \qquad t = \tau_{\bar{R}} \, t', \quad \tau_{\bar{R}} \equiv \frac{\bar{R}^2 \beta}{d_0}. \tag{5}$$

Supersaturation is measured in units of the equilibrium value for a grain of radius $\bar{R}$, velocity in the units that make the kinetic term comparable to the capillary term, and time in units of $\bar{R}$ divided by that velocity.

Substitution. Equation (3) becomes $(d_0/\bar{R})\, u = (d_0/\bar{R})\, \chi' + (d_0/\bar{R})\, v'$. In Eq. (2), $\partial \rho_v / \partial n = \rho_{vs}^I \, (d_0/\bar{R}^2) \, \partial u / \partial n'$, so $v' = \left[D_v \, \beta \, \rho_{vs}^I / (\rho_{\text{ice}} \bar{R}) \right] \partial u / \partial n'$. In Eq. (1) both sides carry $\rho_{vs}^I d_0 / \bar{R}$, leaving $(\bar{R}^2 / D_v \tau_{\bar{R}}) \, \partial u / \partial t' = \nabla'^2 u$ with $\bar{R}^2/(D_v \tau_{\bar{R}}) = d_0/(D_v \beta) \sim 10^{-11}$. The result is

$$\boxed{\nabla'^2 u = 0 \ \text{ in the pore;} \qquad u = \chi' + v', \quad v' = \Lambda \, \frac{\partial u}{\partial n'} \ \text{ on } \Gamma'} \tag{6}$$

$$\Lambda \equiv \frac{L^*}{\bar{R}}, \qquad L^* \equiv \frac{D_v \, \beta \, \rho_{vs}^I}{\rho_{\text{ice}}} = \frac{(1+\delta) \, D_v(T)}{\alpha_c} \sqrt{\frac{2\pi m}{k_B T}}. \tag{7}$$

The saturation vapor density cancels in L^*. The problem has one parameter, Λ, the ratio of attachment to diffusive resistance over a distance $\bar{R}$.

Scaling relation. For a given initial geometry, the surface evolves as a function of $t/\tau_{\bar{R}}$ and Λ only. Hence any quantity Q fixed by the geometry (specific surface area, neck width, $k_{\text{eff}}/k_{\text{ice}}$) satisfies

$$\boxed{Q(t;\, T) = \widehat{Q}\!\left(\frac{t}{\tau_{\bar{R}}(T)};\ \Lambda(T) \right), \qquad \tau_{\bar{R}}(T) = \frac{(1+\delta) \, \bar{R}^2 \rho_{\text{ice}}^2}{\alpha_c \, \gamma} \, \frac{1}{\rho_{vs}^I(T)} \sqrt{\frac{2\pi k_B T}{m}}} \tag{8}$$

where $\tau_{\bar{R}}$ follows from Eqs. (5) and (4). Between two temperatures the time scales are in the ratio

$$a_T \equiv \frac{\tau_{\bar{R}}(T)}{\tau_{\bar{R}}(T_{\text{ref}})} = \sqrt{\frac{T}{T_{\text{ref}}}} \, \frac{\rho_{vs}^I(T_{\text{ref}})}{\rho_{vs}^I(T)}, \tag{9}$$

independent of $\bar{R}$, α_c and γ. With $\rho_{vs}^I \propto \exp(-L_{\rm sub}M_w/R_gT)$ this is an activation energy $E_a = L_{\rm sub}M_w - \frac{1}{2}R_gT \approx 50$ kJ mol^{-1}.

	$-40\,^\circ$C	$-30\,^\circ$C	$-20\,^\circ$C	$-10\,^\circ$C	$-5\,^\circ$C
ρ_{vs}^I [10^{-4} kg m^{-3}]	1.055	3.123	8.487	21.39	33.11
$\tau_{\bar{R}}$ [d]	1747	603	226	91.5	59.7
a_T ($T_{\rm ref} = -20\,^\circ$C)	7.72	2.66	1	0.404	0.264
L^* [μm]	125	132	139	146	150
Λ	2.50	2.64	2.78	2.93	3.00

$\bar{R} = 50$ μm, $\alpha_c = 1.020 \times 10^{-3}$, $\varepsilon = 1$ μm.

From -40 to $-5\,^\circ$C, $\tau_{\bar{R}}$ changes by a factor of 29 while $\Lambda \propto D_v/\sqrt{T} \propto T^{1.31}$ changes by 20%. Neglecting that weak dependence,

$$Q(t;\,T) = Q(t/a_T;\;T_{\rm ref})\,: \tag{10}$$

a change of temperature is a change of the unit of time. Eliminating time between two such quantities gives a relation, such as $k_{\rm eff}$ against SSA, that contains no time scale and depends on temperature only through Λ.

Two time scales. Both have the form $\ell^2\beta/d_0$, the time for a surface of curvature $1/\ell$ to move a distance ℓ. $\tau_{\bar{R}}$ ($\ell = \bar{R}$) is the grain-scale sintering time; $\tau_{\rm sub}$ ($\ell = \varepsilon$) is the relaxation time of the diffuse interface, the parameter the model is written with. $\tau_{\rm sub} = \varepsilon^2\beta/d_0 = (\varepsilon/\bar{R})^2\tau_{\bar{R}}$. With $\varepsilon/\bar{R} = 1/50$ in every run, $\theta = t/\tau_{\rm sub} = 2500\,t/\tau_{\bar{R}}$ and $\tau_{\rm sub}(T)/\tau_{\rm sub}(T_{\rm ref}) = a_T$.

Neglected, with sizes over this range. Vapor storage, $d_0/(\xi_v D_v \beta) \le 2 \times 10^{-8}$. Vapor mass in Eq. (2), $\rho_{vs}^I/(\xi_v\rho_{\rm ice}) \le 4 \times 10^{-3}$. The effect of latent heat on the local saturation,

$$\frac{\rho_{\rm ice}L_{\rm sub}^2\, m\,\bar{R}}{k_BT^2\,k_{\rm ice}\,\beta} \le 2.4 \times 10^{-3}.$$

The relation requires a uniform temperature, a single transport mechanism and an α_c independent of temperature. Numbers: `check_groups.py`.